

*Creating Our Own
In-House*



Mini Tesla

Autonomous Mecanum Drive Robot Platform

Mecanum Drive Train

Full Autonomous Stack

Energy Generation

Sensor Fusion

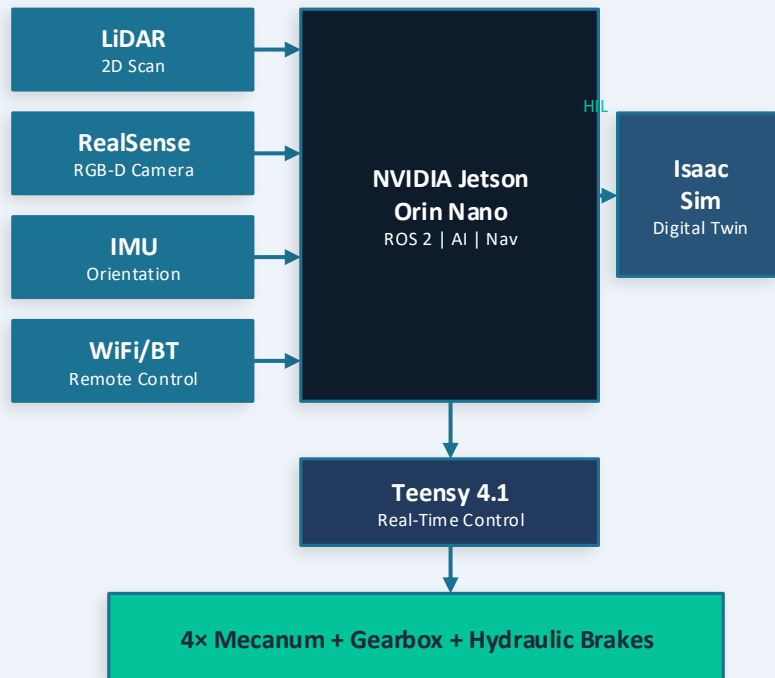
Remote Control

PROJECT OVERVIEW

Project Goal

- Build an omnidirectional mecanum-drive robot — our own in-house Mini Tesla.
- Custom-built gearbox with adaptive gear shifting for incline traversal.
- Design shock absorbers & hydraulic brakes from scratch for real terrain.
- Deploy AI: object detection, human following & AEB emergency braking.
- WiFi/Bluetooth remote control with full ROS 2 autonomous stack.
- Validate via NVIDIA Isaac Sim digital twin before hardware deployment.

System Architecture



KEY CAPABILITIES



Omnidirectional Drive

4-wheel mecanum drive — forward, lateral, diagonal, rotation, all without turning.



AI Perception

Object detection, human following & AEB emergency braking via RealSense + Jetson.



Custom Gearbox

In-house built 2-ratio transmission with servo-actuated selector for inline torque.



Energy Regeneration

Free-wheel motor-as-generator mode redirects kinetic energy back to LiPo battery.



Autonomous Navigation

SLAM + Nav2 path planning with LiDAR localization and dynamic obstacle avoidance.



Remote Control

WiFi & Bluetooth control mode alongside full autonomous stack for manual override.



Digital Twin (Isaac Sim)

Full HIL simulation — validate code in Isaac Sim before touching real hardware.

EDUCATIONAL VALUE

Working on this project develops expertise across six high-demand engineering disciplines:



Embedded Systems

1. Real-time control via Teensy 4.1
2. PWM motor driving & encoder feedback
3. PID velocity loop tuning
4. CAN / serial communication protocols



Mechanical Design

1. Custom gearbox design & fabrication
2. Shock absorber spring-damper physics
3. Hydraulic brake circuit from scratch
4. FEA – Finite Element Analysis



ROS 2 Software

1. Node-based modular architecture
2. SLAM mapping & localization
3. Nav2 autonomous navigation
4. Sensor fusion & odometry



AI & Computer Vision

1. Neural network object detection
2. AEB depth threshold pipeline
3. Person re-identification & following
4. Edge inference on Jetson GPU



Simulation & Testing

1. NVIDIA Isaac Sim 3D setup
2. Hardware-in-the-loop (HIL)
3. Digital twin validation workflow
4. Sim-to-real transfer techniques



Controls & Comms

1. WiFi / Bluetooth remote control
2. Autonomous ↔ manual mode switching
3. ROS 2 teleop integration
4. Emergency stop & failsafe logic

COMPONENT COST ESTIMATE

Component	Category	Qty	Unit	Total
NVIDIA Jetson Orin Nano Super	Compute	1	\$249	\$249
Teensy 4.1 Microcontroller	Embedded	1	\$30	\$30
Pololu 37D Bare DC Motor (24V, Encoder)	Drivetrain	4	\$37.50	\$150
Custom CNC Gearbox (Aluminum, 2-ratio)	Transmission	4	~\$40	~\$160
Servo Motor (Gear Selector)	Transmission	1	\$30	\$30
Cytron MDD10A Dual Motor Driver	Power Elec.	2	\$30	\$60
Intel RealSense D435 Camera	Perception	1	\$399	\$399
RPLIDAR A1 (2D LiDAR)	Perception	1	\$100	\$100
IMU (BNO055)	Sensing	1	\$25	\$25
WiFi / Bluetooth Module	Comms	1	\$25	\$25
LiPo Battery Pack (22.2V 10Ah)	Power	1	\$150	\$150
Power Management / BMS Board	Power	1	\$80	\$80
Robot Chassis (Aluminum Frame)	Structure	1	\$180	\$180
Wiring, Connectors, Misc.	Misc.	—	—	\$120

TOTAL ESTIMATED COST

~\$1,758

** Custom gearbox cost is estimated for materials. ROS 2 and NVIDIA Isaac Sim are free / open-source.*

PROJECT TIMELINE — Spring · Summer · Fall 2026

SPRING 2026

Now → May 13

Procurement — Motors, MCU, Sensors, Battery

CAD Design — Chassis, Gearbox, Brakes, Shockers

FEA Analysis — Structural validation of components

Order & receive all hardware components

M1

MILESTONE #1 — May 13

Chassis physical · Wheels mounted
Brakes CAD · Shockers CAD · Gearbox CAD

SUMMER 2026

May 26 → July 6

3D Print / CNC — Manufacture custom parts

Full Mechanical Assembly — Chassis + all systems

Wiring — MCU, LiDAR, IMU, AI PC, BT/WiFi

Embedded Firmware — Motor control, PID, encoders

Testing — Power regen, gearbox, brakes, remote ctrl

Isaac Sim — Full scale simulation & basic SLAM test

M2

MILESTONE #2 — June 13

Full assembly complete · All wiring done
Gearbox tested · Brakes tested · Remote control · Simulation
Testing

FALL 2026

Aug 24 → Dec 16

Camera Integration — RealSense + SLAM depth layer

AI Models — YOLO object detection, person following

AEB System — Depth threshold emergency braking

Full ROS 2 Stack — Nav2 + sensor fusion on hardware

System Integration — All subsystems working together

Final Demo — Full autonomous run & documentation

M3

MILESTONE #3 — Dec 16 ★

Camera integrated · AI models running
Full autonomous stack on hardware · Final demo